

Introduction to Thermodynamic Properties and Measurement Systems

CHE 260 – Thermodynamics and Heat Transfer



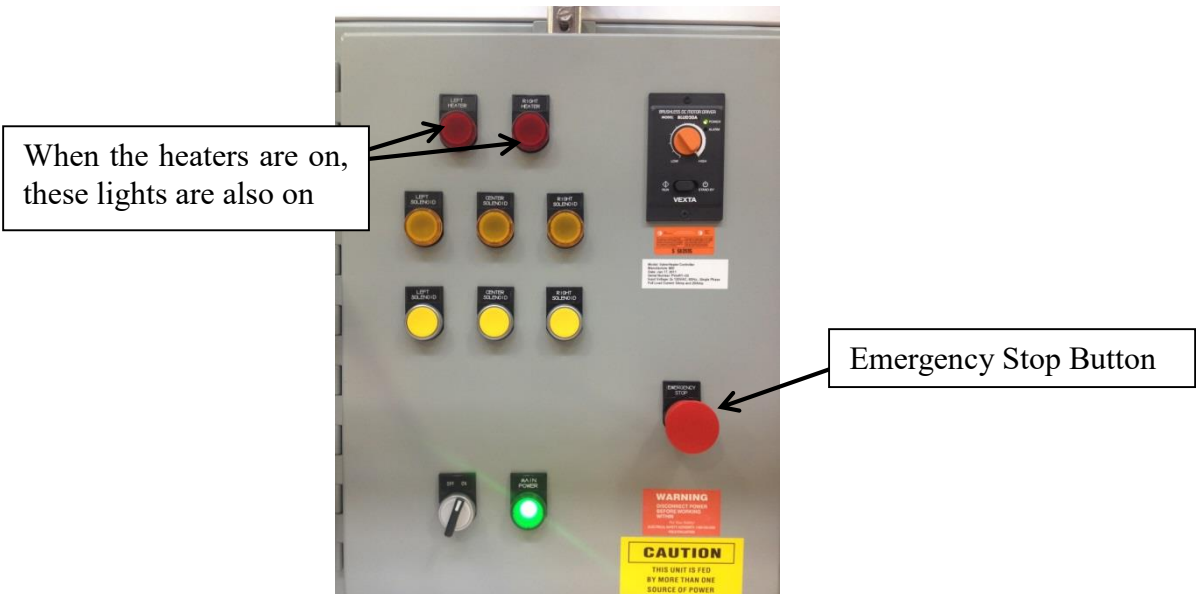
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Safety Warning:

For a safe operation of the system, make sure the following conditions are met:

1. Do not allow the pressure to exceed 80 psi (the regulator pressure will already be preset to 80 psi).
2. Do not allow temperature to exceed 70°C (always keep $T < 70^{\circ}\text{C}$).
3. **Always wear safety glasses when operating this system.**
4. Use the flow meter control in LabVIEW to ramp up the flow of air GRADUALLY from zero. For example, DO NOT set the flow rate to the maximum value and open all the valves to fill a tank or to create a vacuum. Instead, ensuring that the flow rate is set to zero initially, open all the desired valves and then ramp up the flow rate in increments to the maximum flow rate value. This procedure ensures that the flow controllers do not get a large surge of flow through them.
5. When increasing the temperature in labs 1 and 2, if the software does not turn off the heaters after 20 seconds, turn off the electrical control box by pressing the red button 'EMERGENCY STOP' [See image below] and ensure the heater indicating lights are off. Inform your TA immediately.



6. When increasing the temperature, continuously monitor the tank temperature on the graph in LabVIEW; it should never reach 70°C. The tanks are made of acrylic with transition temperature ranging from 85°C to 165°C. Once the acrylic enters this state, it will not be able to hold pressure and will start deforming. In such a case, turn off the electrical control box and ensure the heater indicating lights are off. Inform your TA immediately.

Introduction

Thermodynamics deals with energy and its transformation. However, energy cannot be measured directly, and it is determined using other measurable properties. In engineering thermodynamics, the most common measurable properties are pressure, temperature and density or specific volume. The present experimental facility is designed to introduce various properties and some of the instruments that are used to measure properties.

Lab 1 is aimed at introducing the whole setup and its operation. The same setup will be used in Lab 2 to perform various experiments. Several basic operations are necessary to perform all labs. One operation is filling (increasing the pressure) and emptying (decreasing the pressure) each tank. The other is heating the gases inside the tank and learning how to collect and record the data using LabVIEW software. The ability to fulfill these tasks is of great importance in succeeding labs since the tasks will be used in combination with each other in order to collect the required data.

Pressure

Pressure is a measure of the force exerted per unit area on the boundaries of a system. It is caused by the collisions of the molecules of the substance with the boundaries of the system. As molecules hit the walls, they exert forces that try to push the walls outward. The forces resulting from all these collisions cause the pressure exerted by a system on its surroundings. A pressure gauge [see the image to the right] is an instrument which measures the pressure in a system.

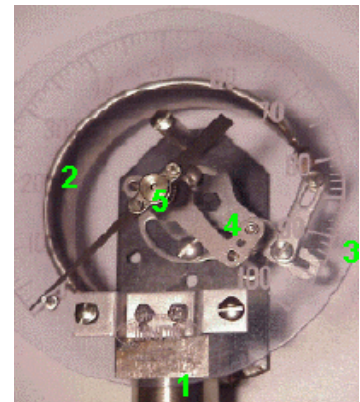


The SI unit for pressure is the Pascal (Pa), equal to one newton per square meter (N/m^2 or $\text{kg}\cdot\text{m}^{-1}\cdot\text{s}^{-2}$). Other commonly used units of pressure are pounds per square inch (psi), *bars*, *atmosphere (atm)*, and *inches of mercury*. The conversion between these units is as follows:

Atmosphere to Pascal:	1 atm = 101 325 Pa $\sim$ 101 kPa $\sim$ 10^5 Pa
Bars to Pascal:	1 bar = 100 kPa = 10^5 Pa
Atmosphere to Bars:	1 atm = 1.01325 bars
Pounds per Square Inch to Pascal:	1 psi = 6895 Pa
Inch of Mercury to Pounds per Square Inch:	1 in. Hg = 0.491 psi

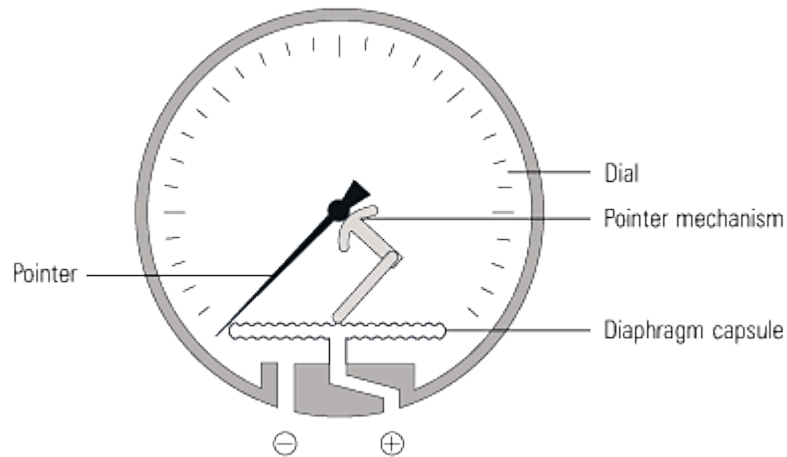
How does a Pressure Gauge Work?

1. **INLET.** Fluid enters through the inlet of the gauge and into area #2, the Bourdon Tube [See image on the next page].
2. **BOURDON TUBE.** The Bourdon Tube is a tubular alloy that has been compressed and bent into an arc. As fluid enters the Bourdon Tube, it causes the tube to flex as it attempts to regain its natural "round" shape. Essentially the arc tries to straighten itself out.
3. **LINKAGE.** The flexure of the Bourdon Tube is linked to the gearing mechanism of the gauge. This is where QUALITY becomes important. As the Bourdon Tube moves, the linkage indexes the gears (#4)



4. **GEARING.** Precision built gears control the movement of the pointer (#5). Finer increments or higher accuracy gauges require more complicated gearing than used by "standard gauges".
5. **POINTER.** The pointer moves with the gearing to indicate the operating pressure within the ANSI accuracy rating of the gauge.

Below is a schematic image of a pressure gauge.



The workstations found in our lab use **WIKAI** pressure gauges. Several limitations of these pressure gauges are:

1. They respond slowly to changes in pressure.
2. They are subjected to hysteresis.
3. They are sensitive to shocks and vibrations.

There are two pressure gauges found at each workstation [see the image below]. The pressure gauge on the left measures the pressure in the left tank whereas the pressure gauge on the right measures the pressure in the right tank.



Manometer

A manometer measures the difference in air or liquid pressure by comparing it to an outside source, usually a sample of Earth's atmosphere. There are several types of manometers (U-tube being the most common). However, the one that is used for the purpose of these labs is a piezometer tube, which consists of a single tube and a base that holds the liquid (in our case, Mercury).

A piezometer is one of the simplest forms of manometers. It can be used for measuring moderate pressures of liquids. The setup of piezometer consists of a glass tube, inserted in the wall of a vessel or of a pipe. The tube extends vertically upward to such a height that liquid can freely rise in it without overflowing. The pressure at any point in the liquid is indicated by the height of the liquid in the tube above that point.

The figure below depicts the manometer which we shall be using to determine the ambient pressure:



Temperature

Temperature is the degree or intensity of heat present in a substance or object. It is expressed according to a comparative scale (Celsius, Fahrenheit, or Kelvin) and is measured using a variety of instruments, generally referred to as thermometers.

Some common scale conversions include:

Celsius to Kelvin: $[K] = [^{\circ}C] + 273.15$

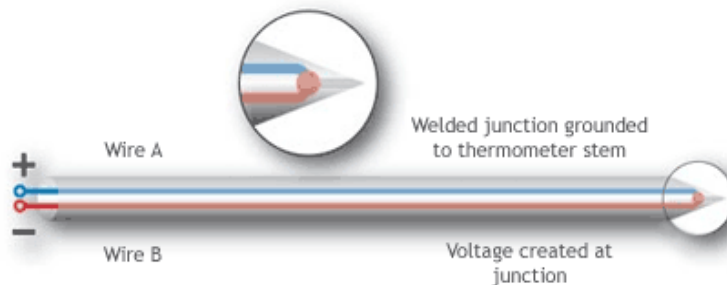
Fahrenheit to Kelvin: $[K] = ([^{\circ}F] + 459.67) \times \frac{5}{9}$

Fahrenheit to Celsius: $[^{\circ}C] = (5/9)([^{\circ}F] - 32)$

Thermocouple

A thermocouple is a sensor for measuring temperature. Thermocouples work based on the principal that when the junction of the two metals is heated or cooled a voltage is produced that can be correlated back to the temperature.

A thermocouple consists of two dissimilar metals (or alloys), joined together at one end with a potentiometer (or a voltmeter) connected in one side of the circuit. An example of these two metal wires can be iron and constantan (an alloy of 60% copper and 40% nickel).



The measuring junction is placed in the environment whose temperature is to be measured. For precision work, the reference junction is kept at a fixed, known temperature (for example, by being placed in an ice bath). If great precision is not required, the reference junction may be left at room temperature, which is known only approximately. The potential difference between the two junctions, as shown by the potentiometer or voltmeter, is used to find the temperature of the heated junction, usually from a table in a handbook.

For the purpose of the MIE210 labs, the thermocouple which will be of primary interest is the one which can be found in the left tank [See Figure 4 below]. This thermocouple is responsible for controlling the temperature which the chamber goes to.

Density and Specific Volume

Density, ρ , is defined as the mass of a substance divided by the volume it occupies:

$$\rho = \frac{\text{Mass}}{\text{Volume}} = \frac{\text{kg}}{\text{m}^3}$$

For the purposes of the current experiments, the density of the gas in each tank is needed. The volume of each tank is fixed. However, the mass inside each tank may be changing. In order to determine the density of the gas in each tank, the mass of the gas in the tank should be determined. The mass inside the tank can be determined by measuring the amount of gas that is put inside the tank. The amount of mass that is put inside the tank can be measured using a flow meter which is installed at the inlet to the system. A flow meter measures either the volumetric flow rate or the mass flow rate of the substance flowing into the system.

Mass Flow Rate: Mass flow rate is the mass of a substance which passes through a given surface per unit of time. Its unit is kilogram per second in SI units. The common symbol used in association with the mass flow rate is $\dot{m}$. Mass flow rate can be related to density, ρ , velocity, V , and flow area, A , as:

$$\dot{m} = \rho VA$$

where VA is referred to as the volumetric flow rate. If the density of the fluid is known, the mass flow rate can be determined by measuring the volumetric flow rate.

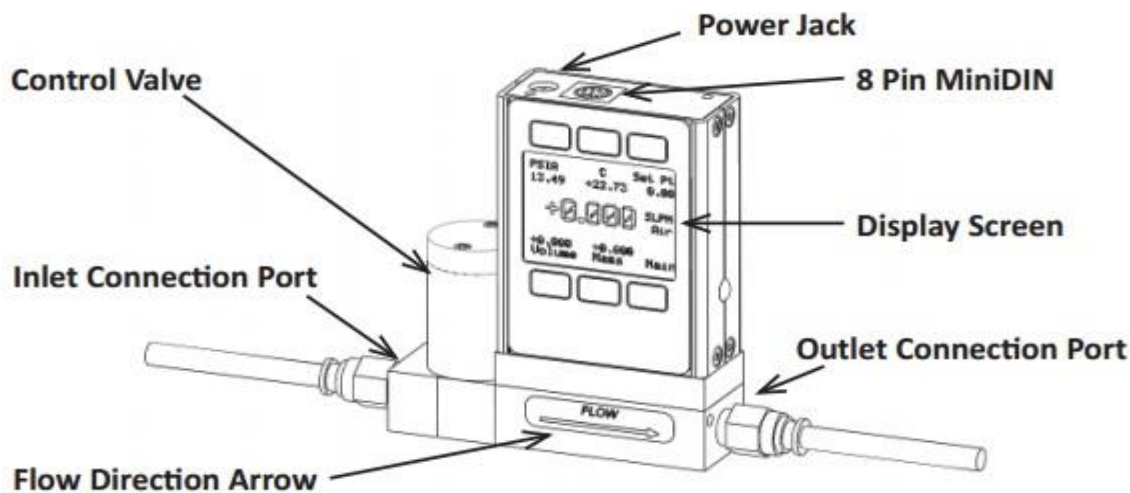
Mass Flow Controller

A mass flow controller (MFC) is a device used to measure and control the flow of gases. A mass flow controller is designed and calibrated to control a specific type of fluid or gas at a particular range of flow rates. The MFC can be given a setpoint from 0 to 100% of its full scale range but is typically operated in the 10 to 90% of full scale where the best accuracy is achieved. The device will then control the rate of flow to the given setpoint. MFCs can be either analog or digital, a digital flow controller is usually able to control more than one type of fluid or gas whereas an analog controller is limited to the fluid for which it was calibrated.

The FMA-2600A Series mass and volumetric flow controllers use the principle of differential pressure within a laminar flow field to determine and control mass flow rate. A laminar flow element (LFE) inside the meter forces the gas into laminar (streamlined) flow. Inside this region, the Poiseuille equation dictates that the volumetric flow rate be linearly related to the pressure drop. A differential pressure sensor is used to measure the pressure drop along a fixed distance of the LFE. This, along with the viscosity of the gas, is used to accurately determine the volumetric flow rate. Separate absolute temperature and pressure sensors are incorporated and correct the volumetric flow rate to a set of standard conditions. This standardized flow rate is commonly

called the mass flow rate and is reported in units such as standard cubic feet per minute (SCFM) or standard liters per minute (SLM).

The following figure is a schematic of the FMA-2600A Series mass and volumetric flow controllers:



PID Controller

A PID controller calculates an "error" value as the difference between a measured process variable and a desired setpoint. The controller attempts to minimize the error by adjusting the process control inputs. PID values determine the performance and operation of your proportional control valve. These terms dictate control speed, control stability, overshoot and oscillation.

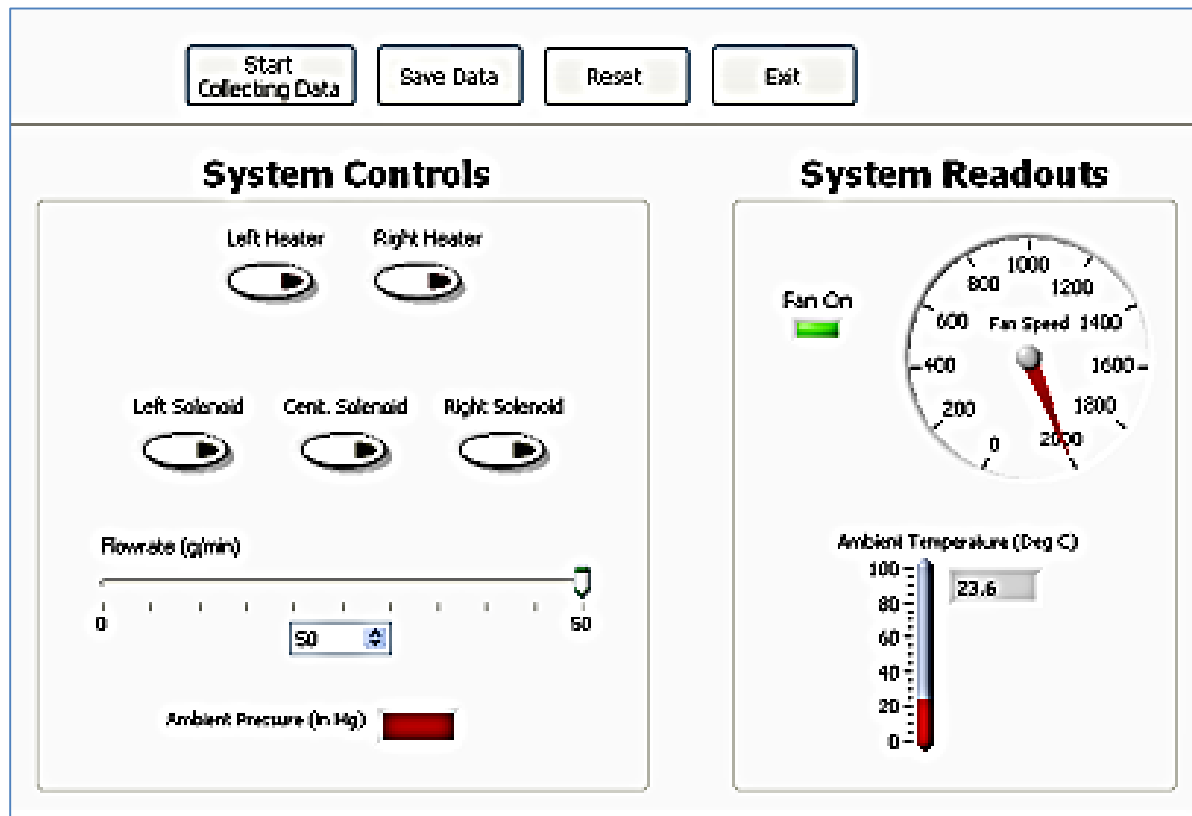
FMA-2600A controllers allow you to adjust the Proportional, Integral and Differential terms of the PID control loop. To change the PID loop parameters, push the button below PID. Press LOOP TYPE. Then use the LN-UP and LN-DN buttons to select the appropriate PID control algorithm. Press SET. Press P, I or D. Then use SELECT to choose the decimal with the arrow and the UP and DOWN buttons to change the value. Press SET to record your value. Press CLEAR to return to zero.

The PD PID algorithm is the PID algorithm used on most FMA-2600A controllers. It is divided into two segments: The first compares the process value to the set-point to generate a proportional error. The proportional error is multiplied by the 'P' gain, with the result added to the output drive register. The second operates on the present process value minus the process value during the immediately previous evaluation cycle. This 'velocity' term is multiplied by the 'D' gain, with the result subtracted from the output drive register. The above additions to and subtractions from the output drive register are carried over from process cycle to process cycle, thus performing the integration function automatically. Increasing the 'P' gain will promote the tendency of the system to overshoot, ring, or oscillate. Increasing the 'D' gain will reduce the tendency of the system to overshoot.

LabVIEW

LabVIEW is a software used for graphical programming and hardware integration to design and deploy measurement and control systems.

The following is a figure of the LabVIEW software's panel controls:

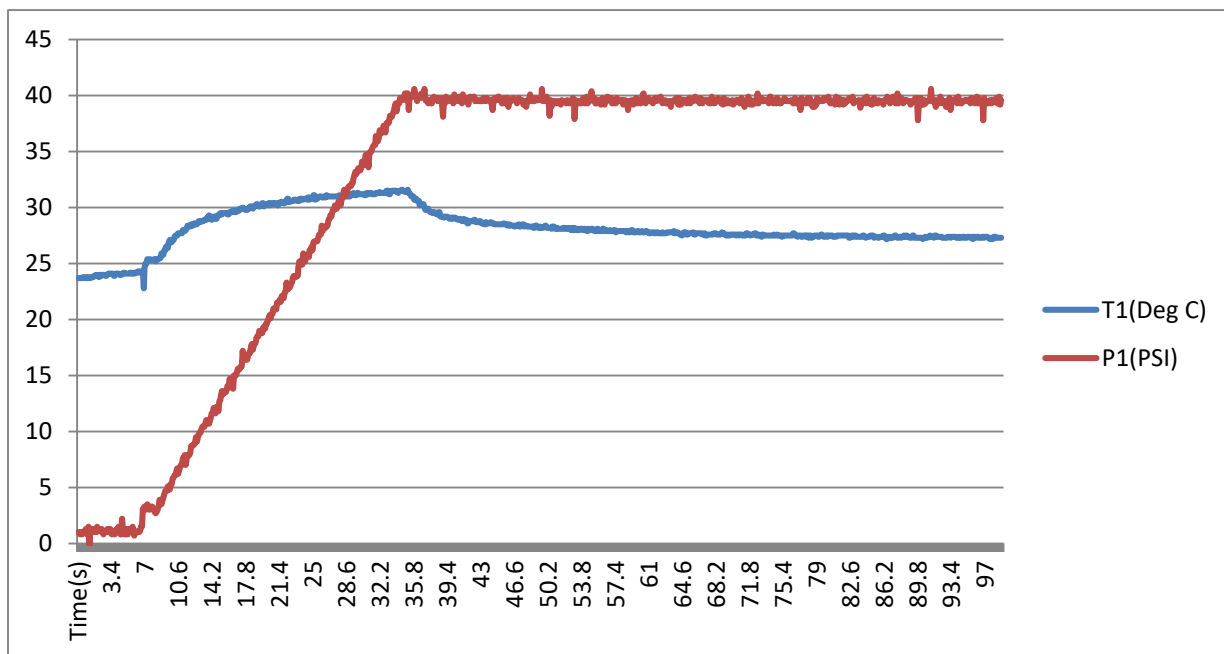


Explanation of Controls

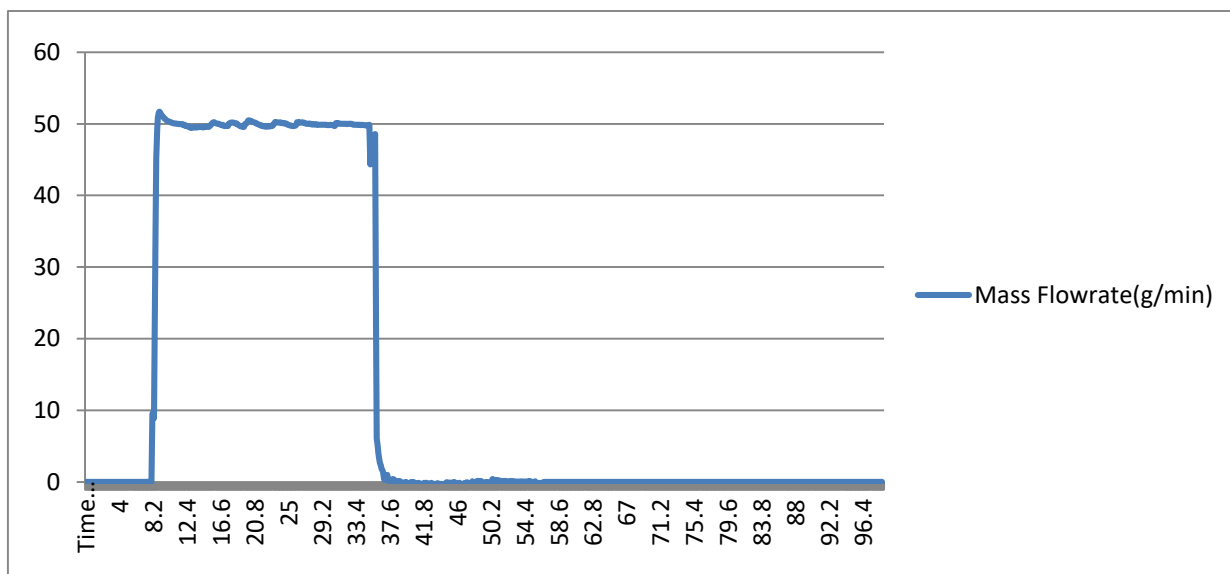
- **Left Heater/Right Heater** – Inputs heat into the left chamber
- **Left Solenoid** – Opens/Closes Left Solenoid (Red=Open, Black=Closed)
- **Center Solenoid** – Opens/Closes Center Solenoid (Red=Open, Black=Closed)
- **Right Solenoid** – Opens/Closes Right Solenoid (Red=Open, Black=Closed)
- **'Start Collecting Data'** – Starts recording results on the graph
- **To set the Flowrate:** Either use the slider OR the text box right below it
- **'Save Data'** – Used to save graph data (you will have to use those numbers provided to make your own graphs)
- **'Reset'** – Clears graphs and all previous results
- **'Exit'** – Closes the Program

Common functions of LabVIEW include measuring and graphing the mass flow rate, temperature and pressure of both the left and right tank, and heater energy.

For example, the following figure depicts the change in temperature and pressure over time in the left tank as it is being pressurized to 40 psig:



Furthermore, the corresponding mass flow rate graph will look like:



As seen above, the mass flow rate is set to 50 g/min, thus allowing gas to enter the left tank which in turn pressurizes it to 40 psig.

Experimental Setup

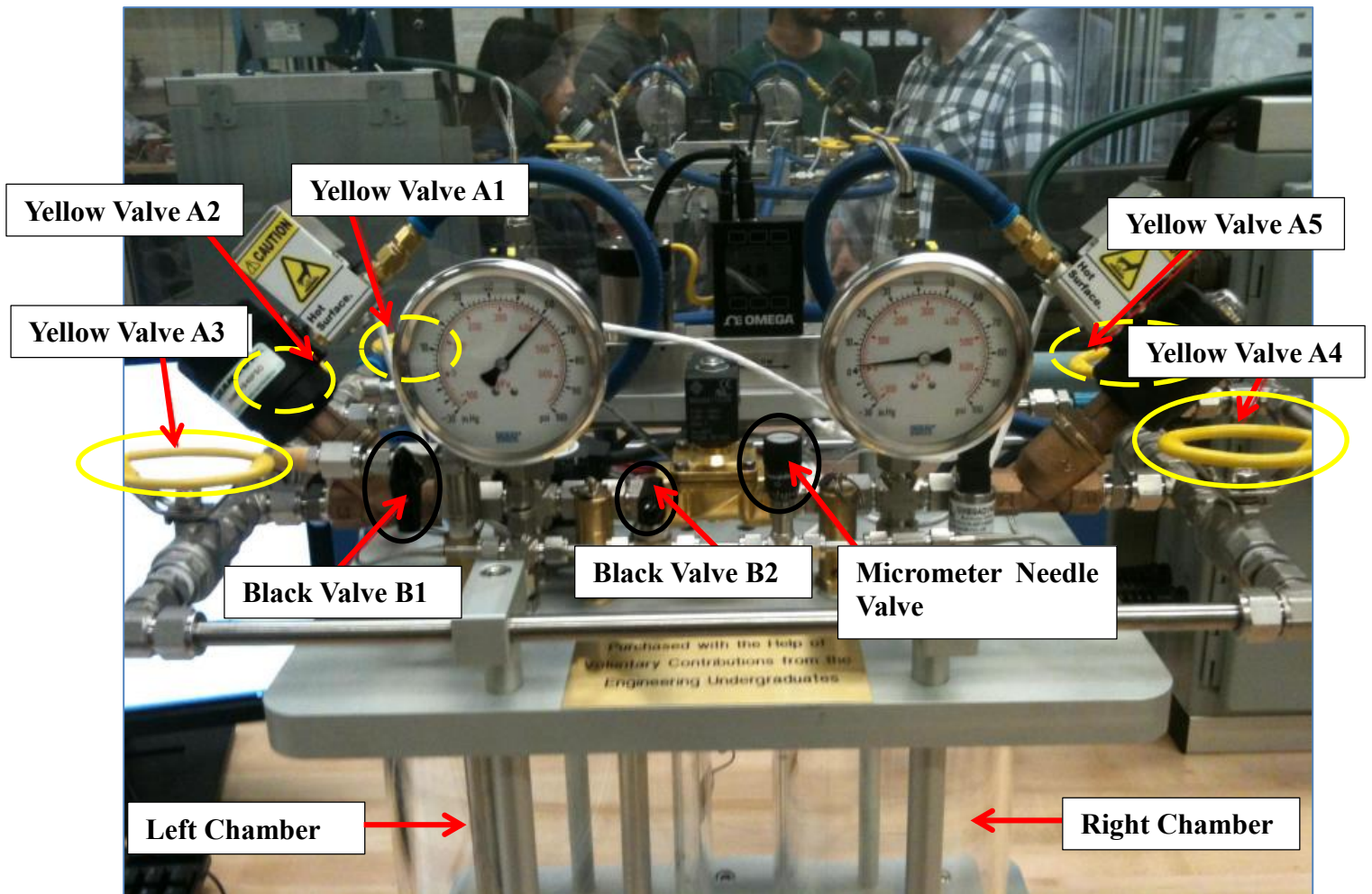


Figure 1: Experimental Setup

***Note: When the black valve B1 is in a vertical position, it means that it is turned OFF.**

In the above figure:

- The yellow valves (A1, A2, A3, A4, and A5) control the flow path of the gas.
- The release valve, Black valve B1, vents the gas out to the atmosphere when opened.
- The micrometer needle valve opens a channel between the left tank and the right tank, thus allowing gas to transfer into the right tank from the left tank. The more the micrometer needle valve is opened, the faster the rate of gas transfer.

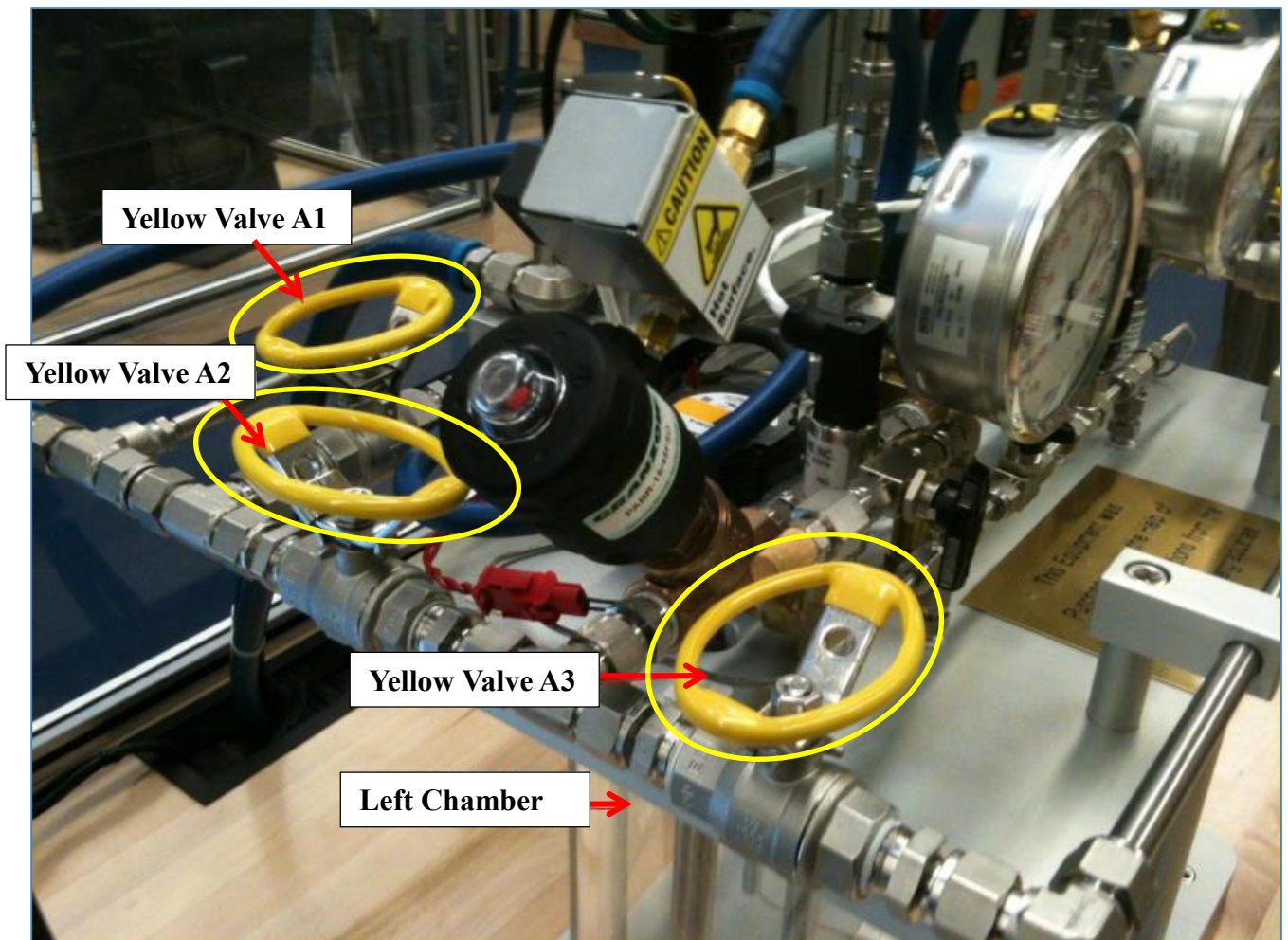


Figure 2: Left Side View – Manually-Controlled Valves

***Note:** When a yellow valve is in the perpendicular to the pipe, it means that it is turned OFF.

****Note:** In the figure above, yellow valve A2 is in a parallel position to the pipe; hence, it is turned ON.

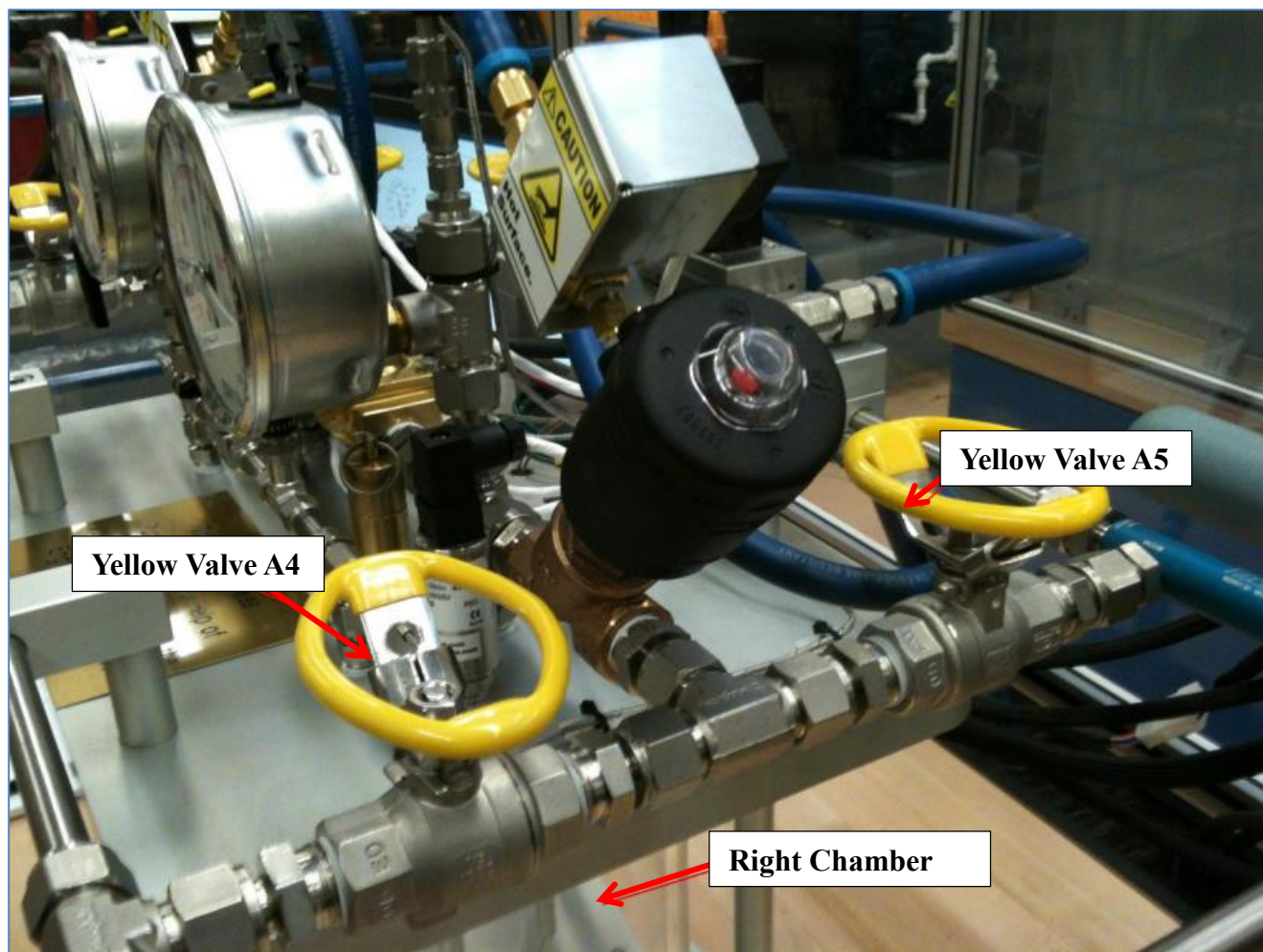


Figure 4: Right Side View – Manually-Controlled Valves

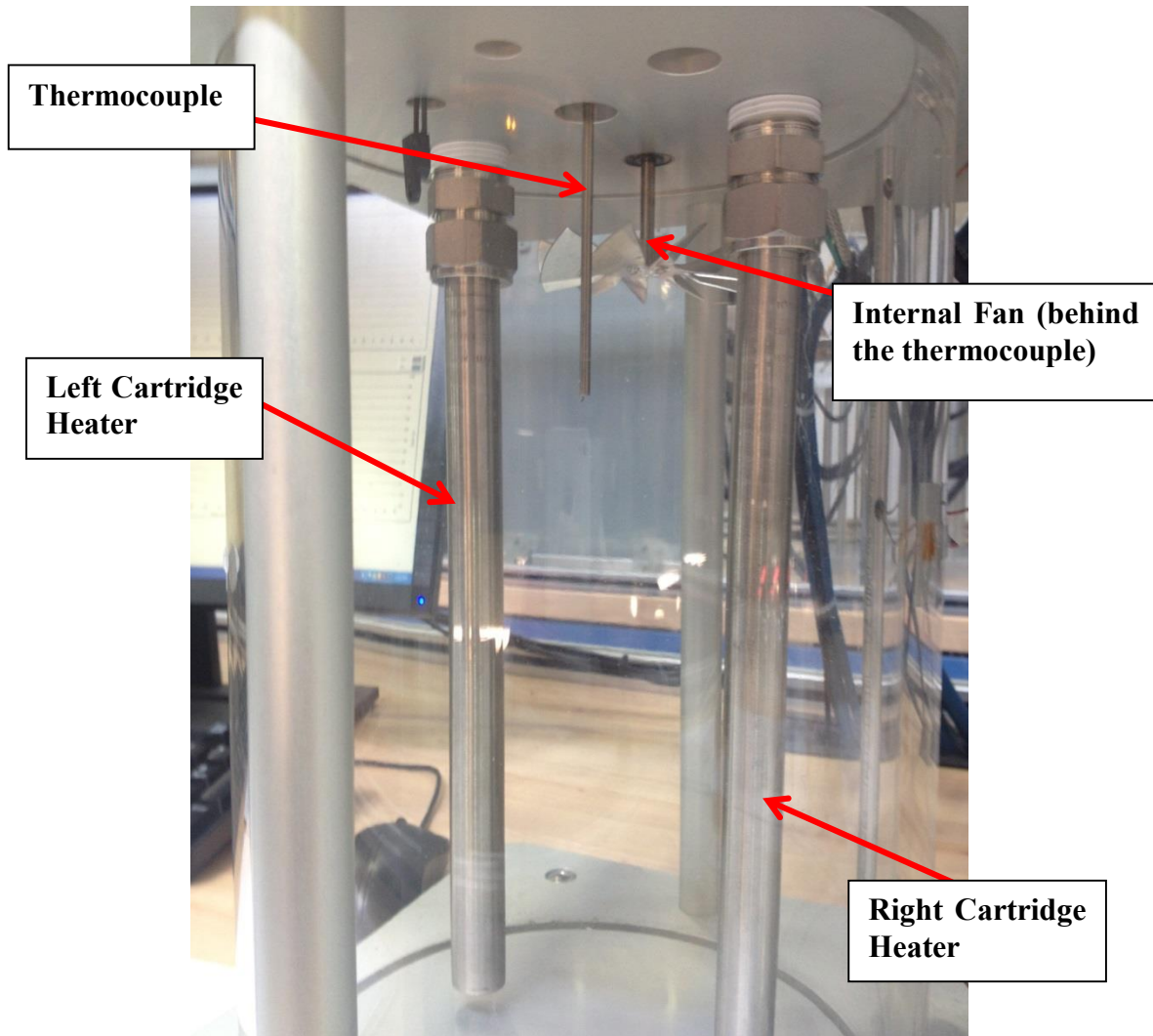


Figure 5: Left Chamber

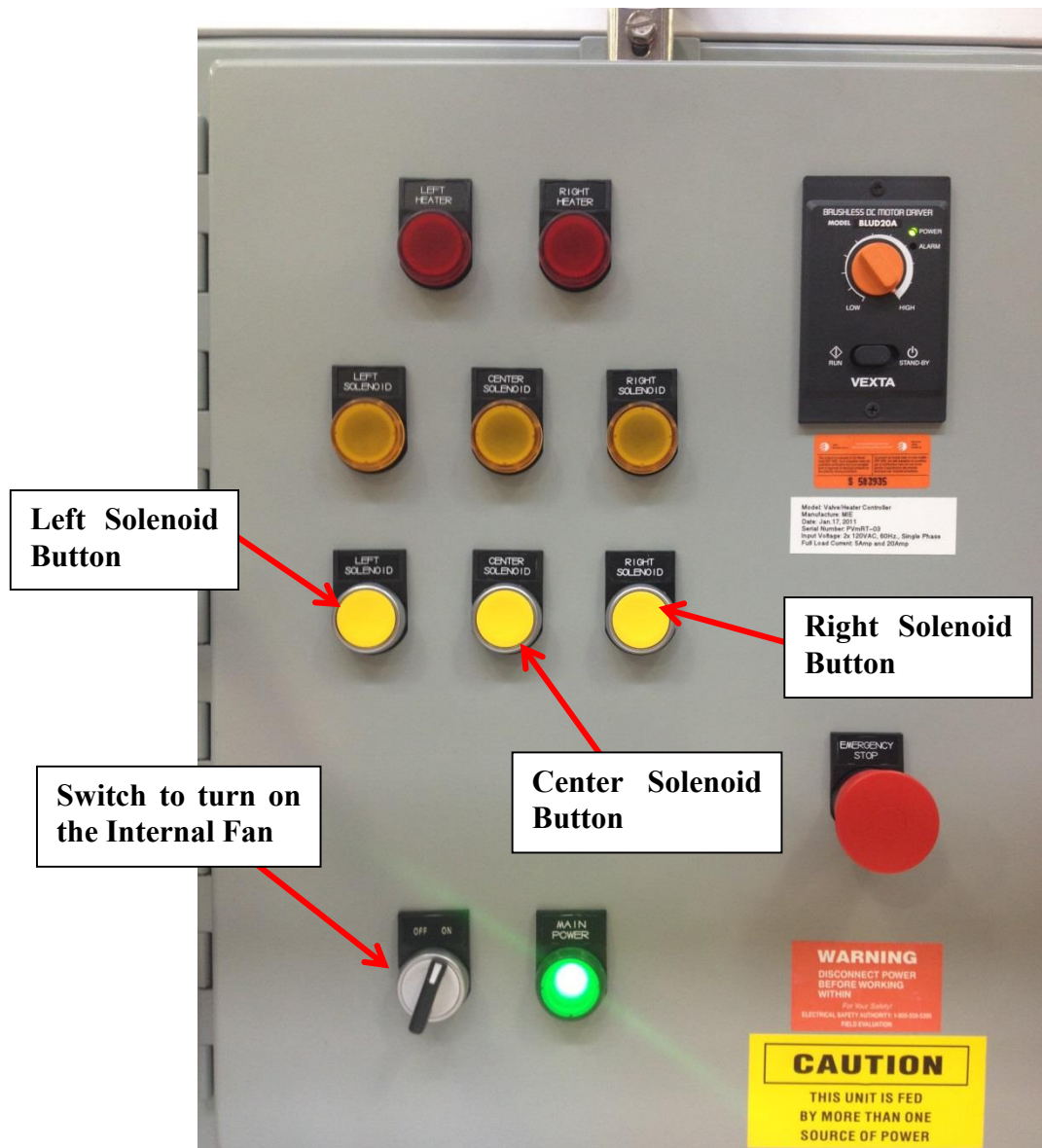


Figure 6: Control Panel

***NOTE: The Internal Fan must be turned ON before data can be collected.**

****NOTE: To manually keep a solenoid open, the button must remain pushed in.**

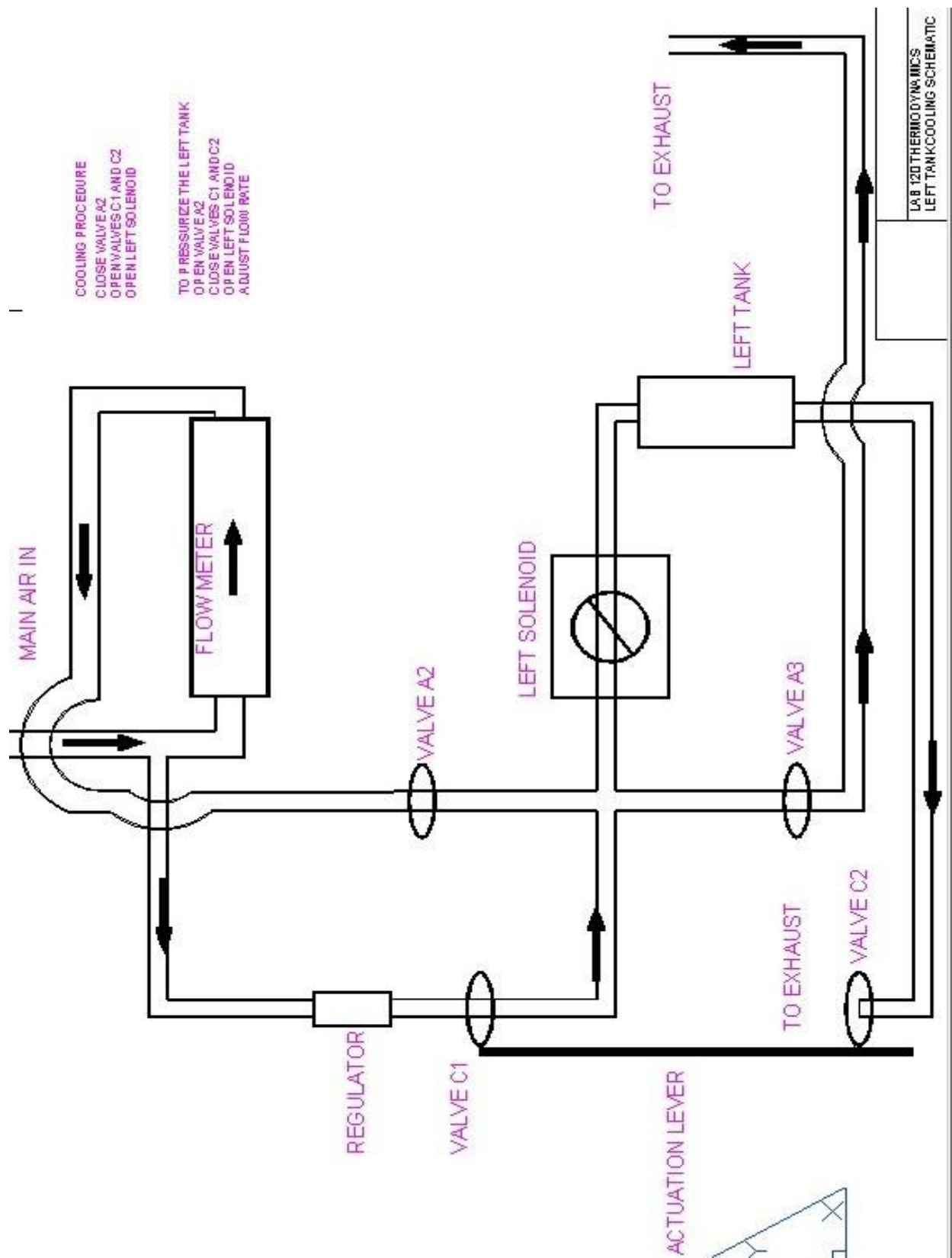


Figure 8: Airflow Schematic of Cooling System

Experimental Procedure

General

To begin any lab:

1. Turn the computer on and start Labview (or click 'Reset' button if Labview is already running)
2. Make sure all valves are closed.

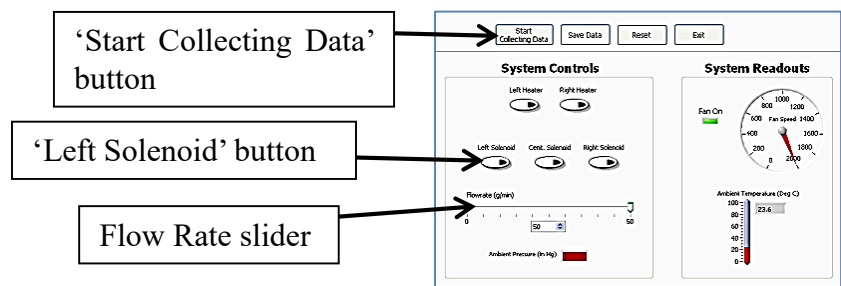
To turn on Labview (if it's not already running):

1. On the desktop, find the folder entitled 'MIE210 – Labs' and open it.
2. Find and open the file corresponding to the lab you're working on (in this case, MIE210_Lab1).

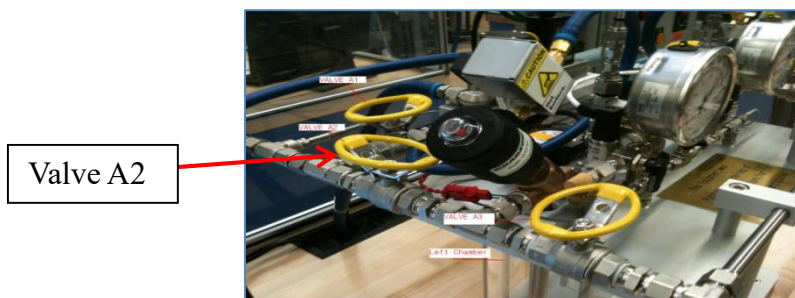
Lesson 1 – How to Set the Pressure in the Left Chamber

Method 1 – Automatic

1. Click the 'Start Collecting Data' button in LabVIEW to begin recording the results on the graphs.



2. Open Valve A2.

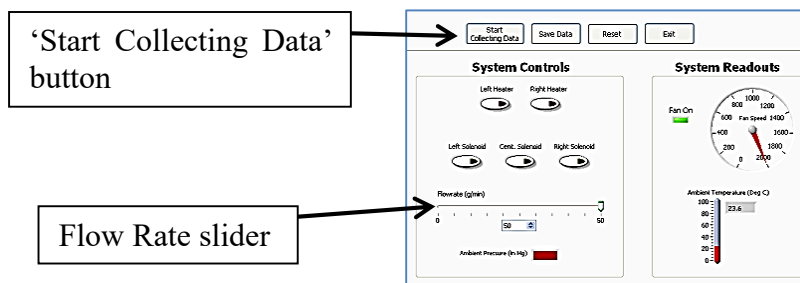


3. Open the Left Solenoid valve by pressing the 'Left Solenoid' button in Labview once.
4. Set the Flow Rate to 50 g/min
5. When the pressure reaches 50 psi in the left chamber, click the 'Left Solenoid' button in the program once.
6. Set the Flow Rate back to 0 g/min.

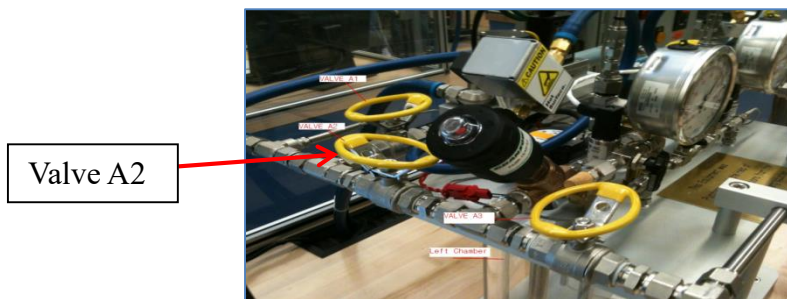
Method 2 – Manual

**** (FYI – If you’ve conducted Method 1, you’ll need to first empty the tank [Lesson 2] before trying this method) ****

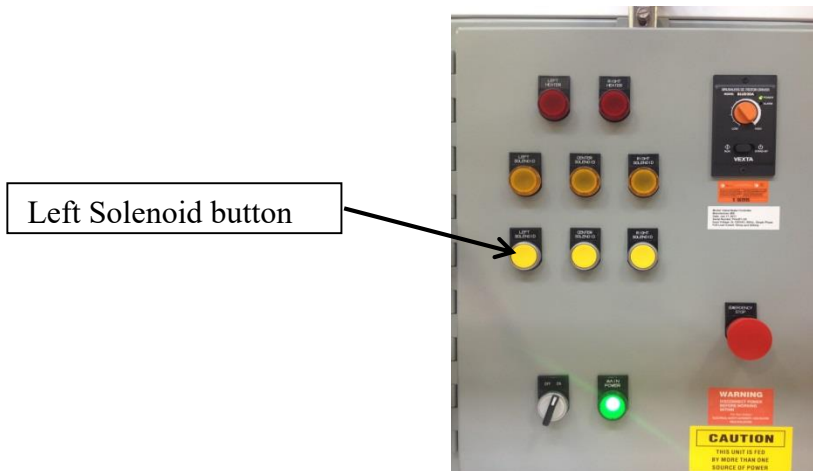
1. Click the ‘Start Collecting Data’ button in the program to begin recording the results on the graphs.



2. Open Valve A2.



3. Open the Left Solenoid valve by pressing and holding the button on the control board.

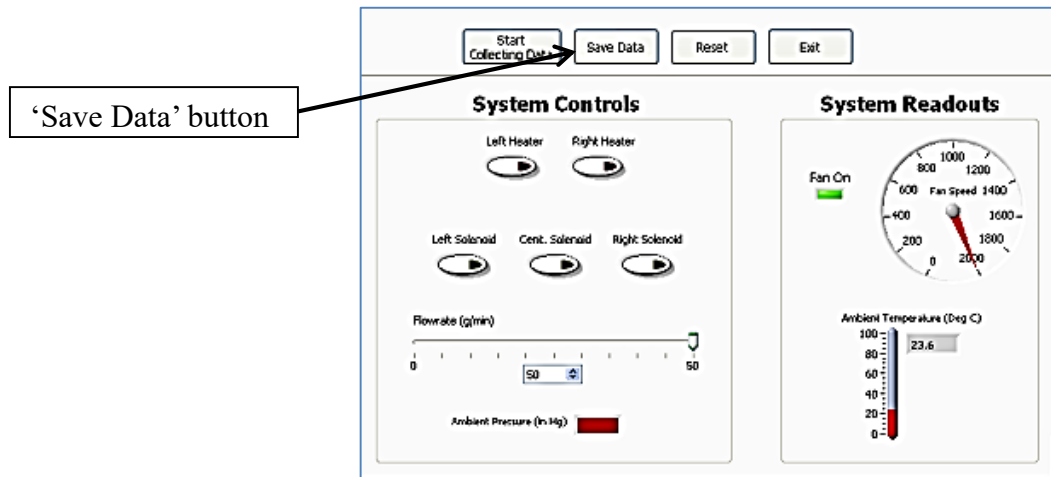


4. Set the Flow Rate to 50 g/min.
5. When the pressure reaches 50 psi in the left chamber, release the ‘Left Solenoid’ button.
6. Set the Flow Rate back to 0 g/min.

After Lesson 1 has been completed using Method 1, the ‘Stop Recording’ button should be clicked in the program to stop recording the results on the graphs.

To save your graphs:

1. Once you're done collecting data, click the 'Save Data' button in LabVIEW.



2. Since this is Lab #1, create a folder on the desktop with your group's name.
3. Save all necessary data in the folder you created in step #2.

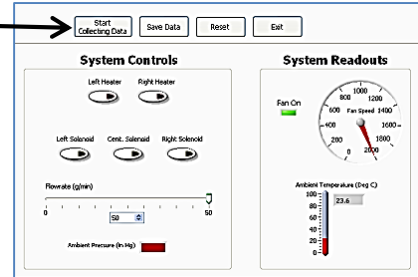
NOTE: Please use the same workstation for all the following labs in order to have access to your group's folder and files.

Lesson 2 – How to Empty the Left Chamber

Method 1 - Manual

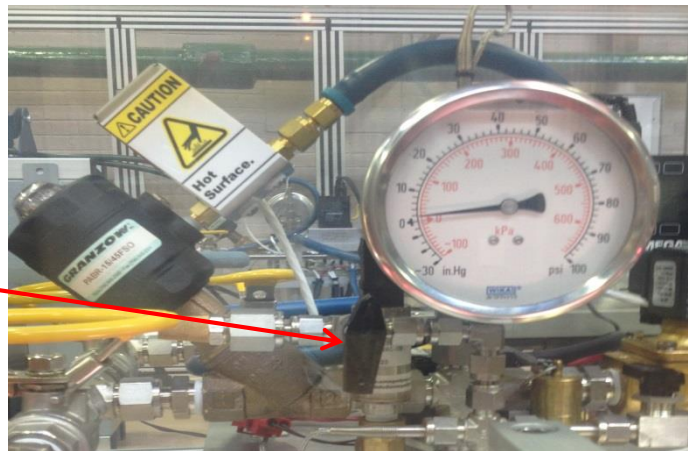
1. Click the 'Start Collecting Data' in the program to begin recording the results on the graphs.

'Start Collecting Data' button



2. Open the black pressure release valve (Valve B1).

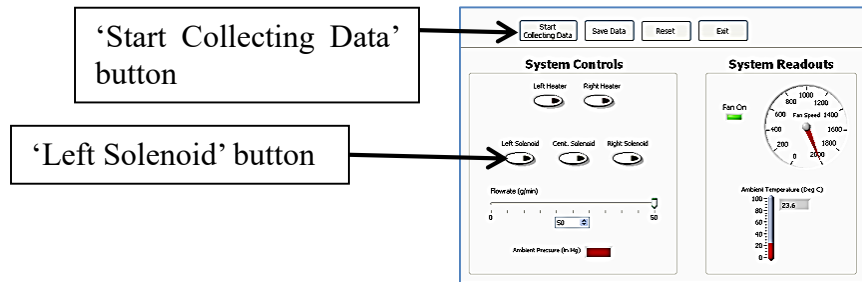
Valve B1



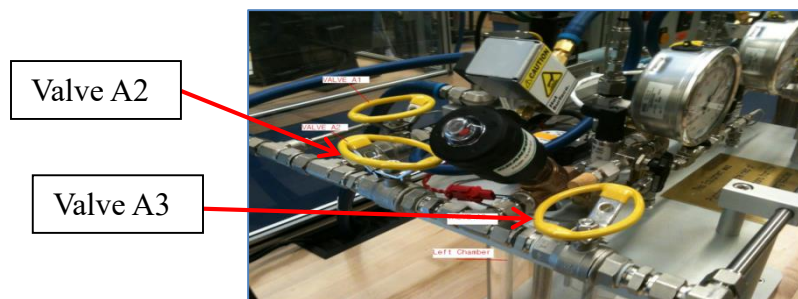
3. When the pressure reaches 0 psi, close the black pressure release valve (Valve B1).

Method 2 - Automatic

1. Click the 'Start Collecting Data' in the program to begin recording the results on the graphs.



2. Close valve A2 and open valve A3



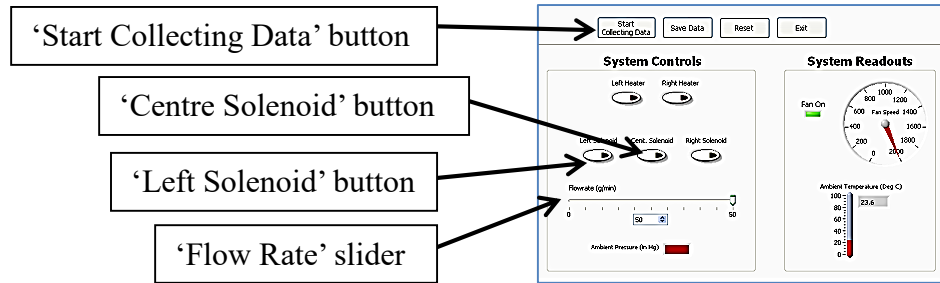
3. Open the Left Solenoid valve by pressing the 'Left Solenoid' button in the program once
4. When the pressure reaches 0 psi close the Left Solenoid valve by pressing the 'Left Solenoid' button in Labview once and close valve A3.

Lesson 3 – How to Increase Pressure in the Left Chamber and Right Chamber

This lesson will teach you how to increase the pressure in the left and right chambers to 35 psi and 25 psi, respectively.

Method 1

1. Click the 'Start Collecting Data' button in the program to begin recording the results on the graphs.



2. Open valve A2.



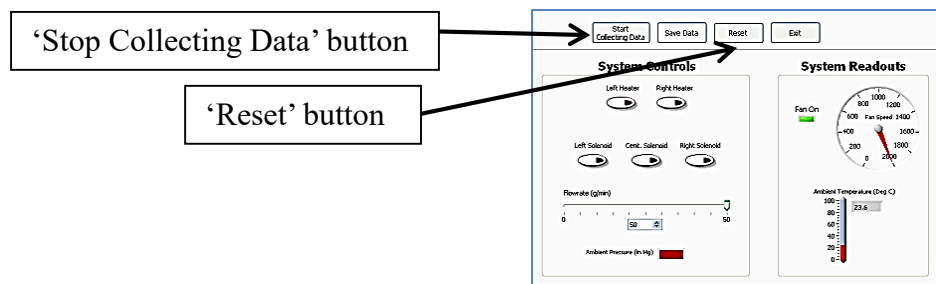
3. Click the 'Left Solenoid' button in Labview to open the left solenoid valve.
4. Open the center solenoid valve by clicking the 'Centre Solenoid' button in the program.
5. Set the Flow Rate to 50 g/min. The pressure in both chambers should be increasing at roughly the same rate.
6. When the pressure in the right chamber reaches 25 psi, close the center solenoid valve by clicking the 'Centre Solenoid' button in LabVIEW. Closing this valve prevents air from entering and exiting the right chamber and thus the pressure of 25 psi is maintained.
7. The pressure of the left chamber should continue to increase. When the pressure in the left chamber reached 35 psi, close the left solenoid valve by clicking the 'Left Solenoid' button on the computer. The left and right chambers have pressures of 35 psi and 25 psi, respectively.
8. Set the Flow Rate back to 0 g/min.
9. Close valve A2.
10. Bring the pressure in both chambers to 0 psi by opening the black pressure release valve (B1) and the center solenoid valve by clicking the 'Centre Solenoid' button on the computer.

Method 1 continues on the next page



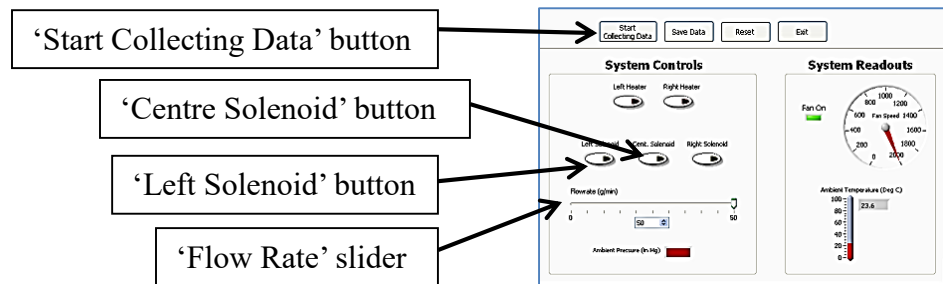
11. Close the black pressure release valve (B1) and close the center solenoid by clicking the ‘Centre Solenoid’ button in Labview.

After Method 1 is completed, click ‘Stop Recording’, followed by clicking ‘Reset’.

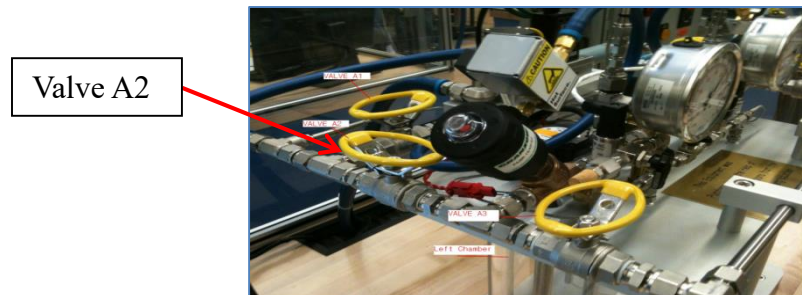


Method 2

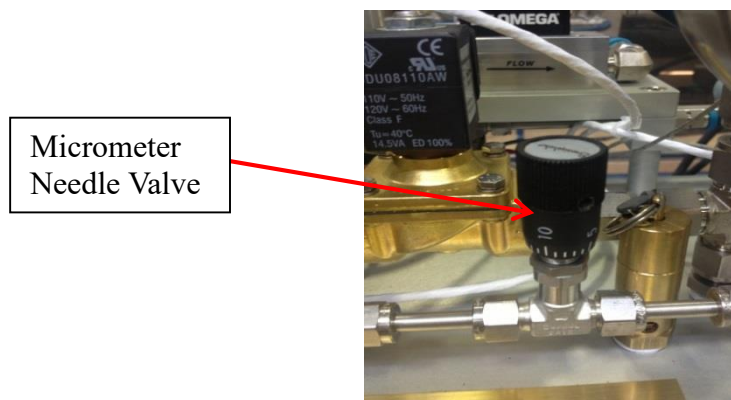
1. Click the 'Start Collecting Data' button in the program to begin recording the results on the graphs.



2. Open valve A2.



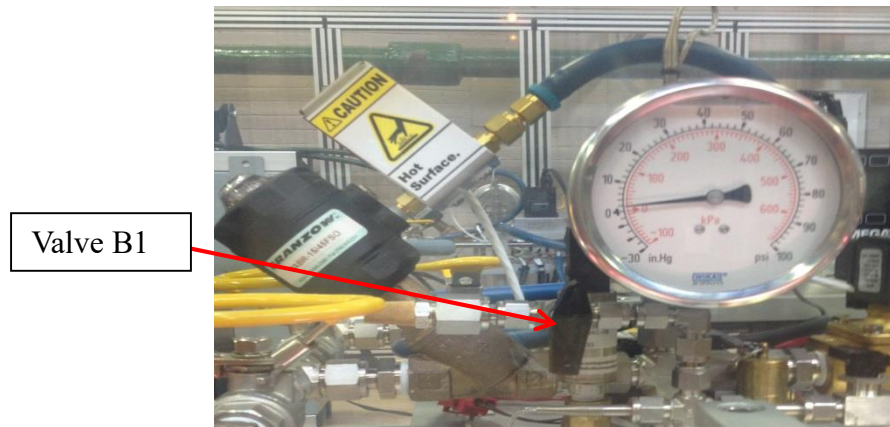
3. Open the left solenoid valve by clicking the 'Left Solenoid' button in Labview.
4. Set the Flow Rate to 50 g/min.
5. When the pressure reaches 65 psi in the left chamber, close the left solenoid valve by clicking the 'Left Solenoid' button on the computer. Air now stops flowing into the left chamber.
6. Set the Flow Rate back to 0 g/min.
7. Close valve A2.
8. Open the micrometer needle valve to allow the air to flow from left to right chamber.



9. When the pressure in the right chamber reaches close to 23 psi, close the micrometer needle valve gradually to slow the amount of air flowing into the right chamber and completely close it when the pressure in the right chamber reaches 25 psi.

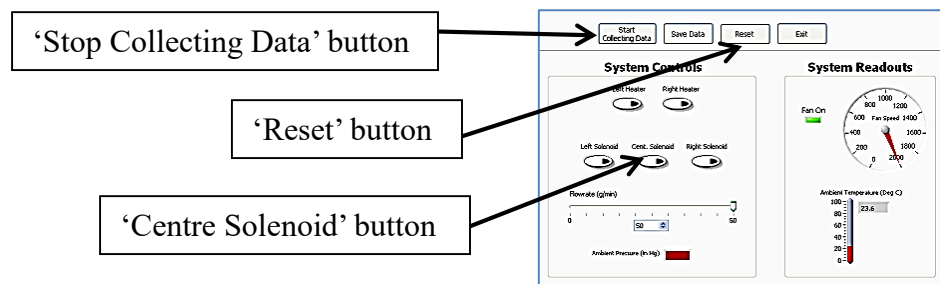
Method 2 continues on the next page

10. Open the black pressure release valve (B1) to bring the pressure down to 35 psi.



11. When the pressure in the left chamber is 35 psi, close the black pressure release valve.

12. Bring the pressure in both chambers to 0 psi by opening the black pressure release valve (B1) and the center solenoid valve by clicking the 'Centre Solenoid' button on the computer.

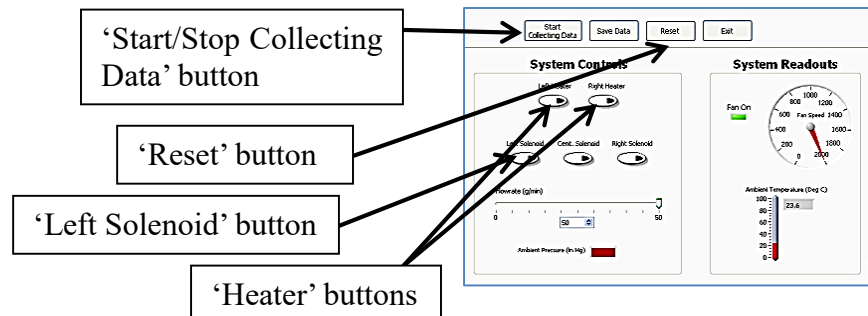


13. Close the black pressure release valve (B1) and close the center solenoid by clicking the 'Centre Solenoid' button on the computer.

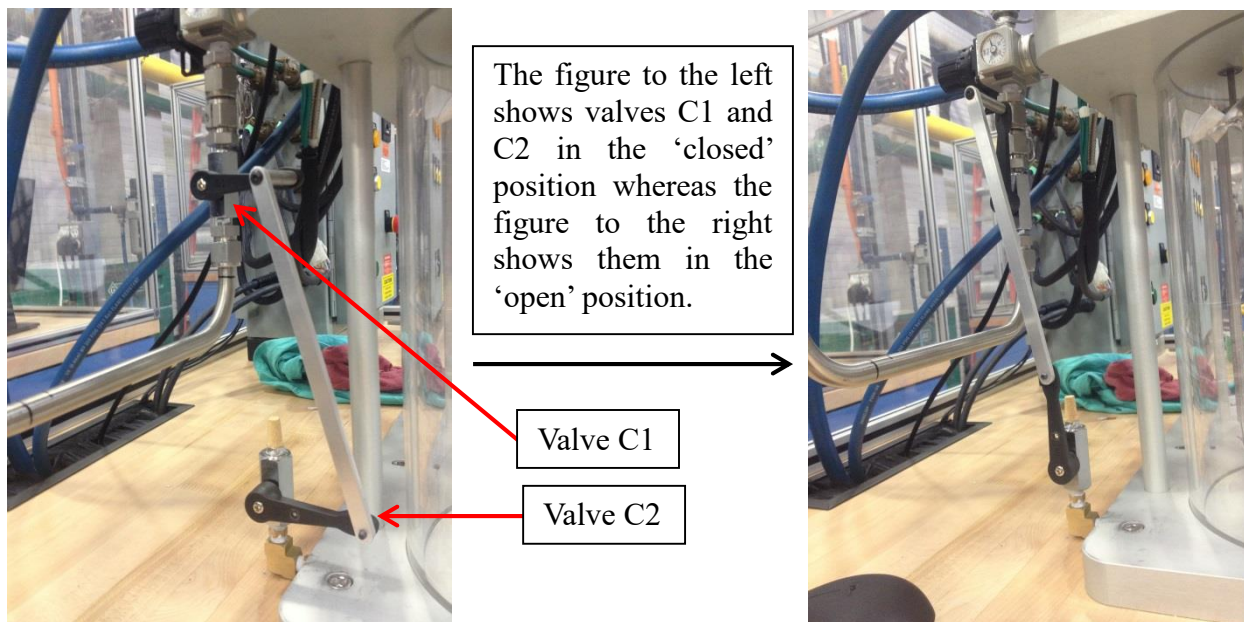
After Method 2 is completed, click 'Stop Recording', followed by clicking 'Reset'.

Lesson 4 – How to Increase the Temperature in the Left Chamber

1. Click the 'Start Collecting Data' button in the program to begin recording the results on the graphs.



2. To increase the temperature, turn on the heaters by clicking either the 'Left Heater' or 'Right Heater' button once, wait 15 seconds, then turn off the heaters by clicking either the 'Left Heater' or 'Right Heater' button once again. **NOTE:** Clicking either one of the heater buttons once in the LABVIEW software will send power to both heaters (both red lights light up on the electrical box) for 15 seconds.
3. To cool the left tank back to ambient temperature, do the following:
 - a. Open valves C1 and C2.
 - b. Open the left solenoid by clicking the 'Left Solenoid' button in LabVIEW and keep it open for several minutes.



4. Once the temperature in the left tank returns to ambient temperature, close the left solenoid and valves C1 and C2.

After Lesson 4 is completed, click 'Stop Recording', followed by clicking 'Reset'.